Umbrella 4: Plant Anatomy & Embryology

Plant anatomy and embryology together reveal the remarkable structural complexity and reproductive sophistication of the plant kingdom. From the microscopic organization of tissues to the intricate choreography of double fertilization, these topics form the backbone of our understanding of how plants grow, develop, and reproduce. This document provides comprehensive notes on tissue systems, meristems, secondary growth, vascular tissues, wood anatomy, anomalous growth patterns, and the complete reproductive cycle from spore formation to seed structure.

BOTANY PLANT ANATOMY EMBRYOLOGY REPRODUCTIVE BIOLOGY

Tissue Systems

The plant body is organized into three fundamental tissue systems, each serving distinct structural and physiological roles. These systems are continuous throughout the plant, from roots to leaves, and are modified according to the specific functional demands of each organ. Understanding tissue systems is foundational to all of plant anatomy, as every organ and structure can be understood in terms of these three systems and their cellular components.

Epidermal Tissue System

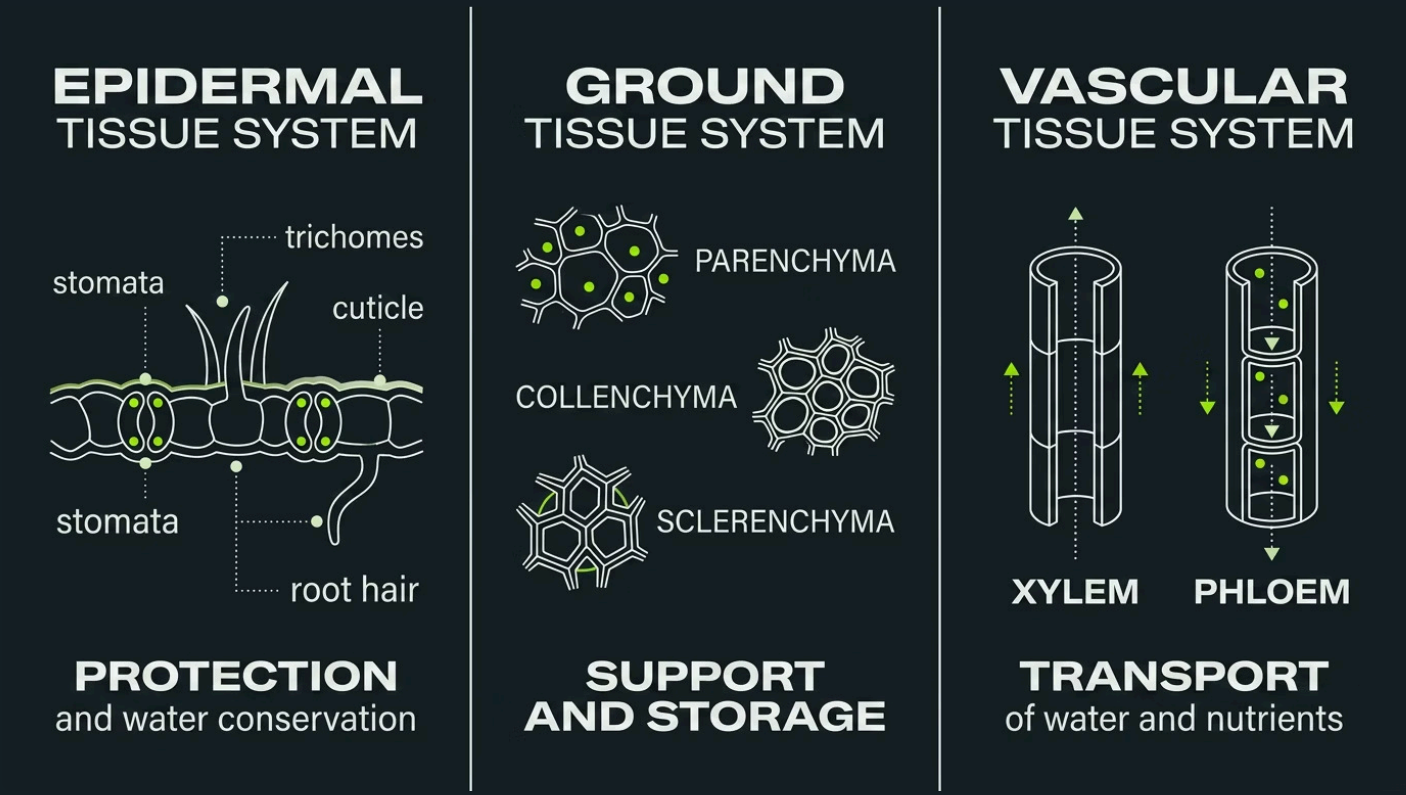
The outermost protective covering of the primary plant body. It includes **stomata** (pores for gas exchange regulated by guard cells), **trichomes** (hair-like outgrowths that reduce water loss and deter herbivores), and **root hairs** (tubular extensions that dramatically increase absorptive surface area). The epidermis is typically a single cell layer covered by a waxy cuticle that prevents desiccation. In woody plants, the epidermis is eventually replaced by periderm during secondary growth.

Ground Tissue System

The fundamental tissue that fills the interior of the plant body between the epidermis and vascular tissues. It comprises three cell types: **Parenchyma** — thin-walled, living cells involved in photosynthesis, storage, and secretion; **Collenchyma** — elongated cells with unevenly thickened primary walls providing flexible support in growing regions; and **Sclerenchyma** — cells with thick, lignified secondary walls (fibers and sclereids) providing rigid structural support. Ground tissue is differentiated into cortex and pith in stems, and cortex in roots.

Vascular Tissue System

The conducting system responsible for long-distance transport of water, minerals, and organic nutrients. It consists of **xylem** (conducts water and dissolved minerals upward from roots) and **phloem** (transports photosynthates and signaling molecules bidirectionally). Vascular tissues are organized into bundles — in stems, these may be arranged in a ring (dicots) or scattered (monocots); in roots, they form a central stele. Vascular bundles may be open (with cambium) or closed (without cambium).



Meristems

Meristems are regions of active cell division that drive all plant growth and development. Unlike animals, plants retain embryonic, undifferentiated cells throughout their lives, allowing for continuous growth and regeneration. Meristems are classified by their location within the plant body and by their origin. The cells within meristems are typically small, isodiametric, with dense cytoplasm, large nuclei, and thin cell walls — characteristics that reflect their high metabolic activity and rapid division rate.

Classification by Location

1 Apical Meristems

Located at the tips of roots and shoots. Responsible for **primary growth** — the increase in length of the plant axis. Root apical meristem is protected by the root cap; shoot apical meristem gives rise to leaves and lateral buds.

2 Intercalary Meristems

Found at the bases of internodes or leaf sheaths, especially in grasses and monocots. Allow rapid regrowth after grazing or mowing. Contribute to elongation of already-formed organs.

3 Lateral Meristems

Include the **vascular cambium** and **cork cambium (phellogen)**. Responsible for **secondary growth** — the increase in girth or diameter of stems and roots. Produce secondary vascular tissues and periderm respectively.

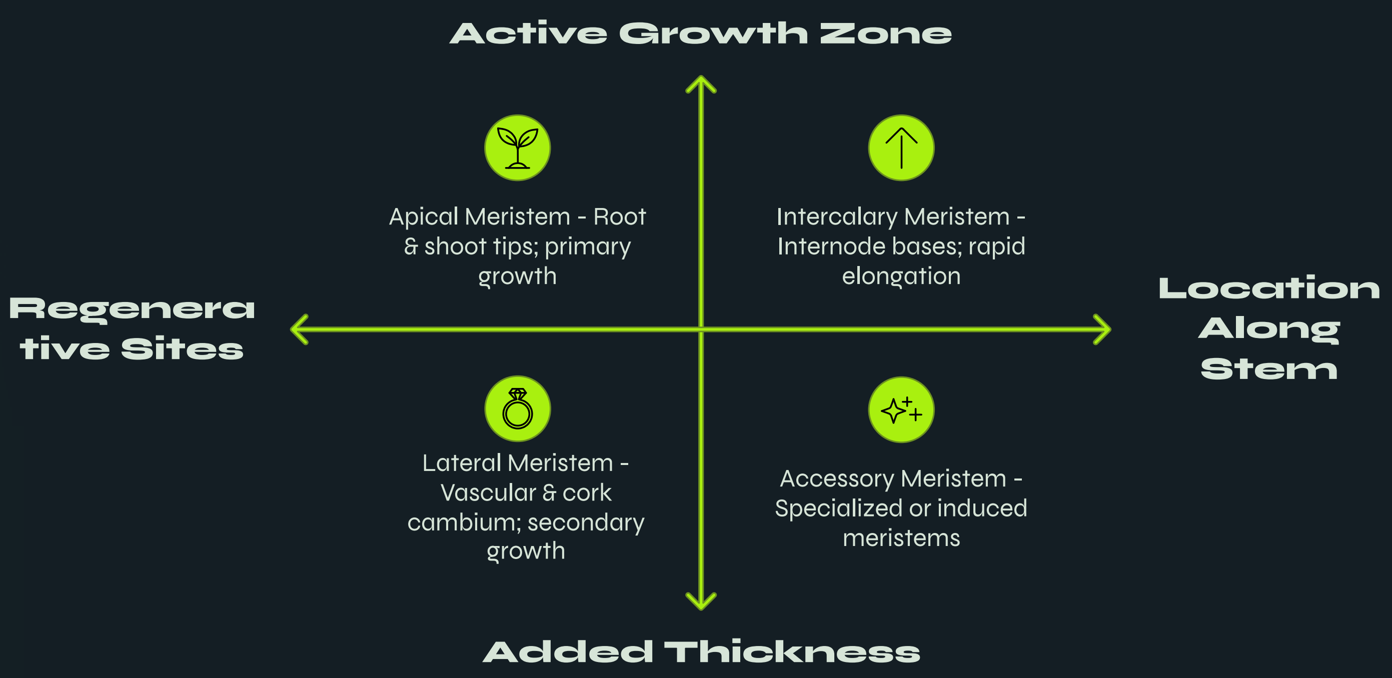
Classification by Origin

Meristems can also be classified based on their developmental origin: **Promeristem** (primordial meristem from embryo), **Primary meristem** (derived from promeristem, includes protoderm, ground meristem, and procambium), and **Secondary meristem** (formed from permanent tissues, e.g., cork cambium, interfascicular cambium).

The Shoot Apical Meristem (SAM)

The SAM is organized into distinct zones: the **central zone** (stem cell reservoir), the **peripheral zone** (where lateral organs are initiated), and the **rib zone** (which contributes to stem tissue). The **tunica-corpus theory** describes the SAM as having an outer tunica (surface layers dividing anticlinally) and an inner corpus (mass of cells dividing in all planes).

The **quiescent center** in root apical meristems is a region of slowly dividing cells that serves as a reserve of stem cells and is critical for meristem maintenance and regeneration after damage.



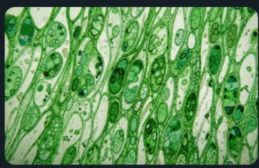
The three meristem types work in concert throughout a plant's life — apical meristems extend the plant body, intercalary meristems enable rapid regrowth, and lateral meristems thicken stems and roots for structural support.

Secondary Growth, Wood Anatomy, Xylem & Phloem

Secondary growth is the increase in girth of stems and roots through the activity of lateral meristems. This process is characteristic of gymnosperms and dicotyledonous angiosperms, and is responsible for the formation of wood and bark. The two lateral meristems involved are the **vascular cambium** (producing secondary xylem and phloem) and the **cork cambium or phellogen** (producing periderm). Together, all tissues outside the vascular cambium constitute the bark.

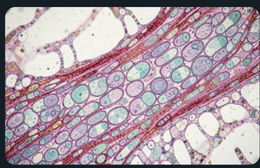
Secondary Growth & Wood Anatomy

Secondary xylem (wood) accumulates in concentric layers, forming **annual rings** in temperate climates — each ring representing one growing season. **Spring wood (early wood)** has wider vessels and thinner walls; **autumn wood (late wood)** has narrower vessels and thicker walls. **Heartwood** is the dark, non-conducting central region filled with resins and tannins; **sapwood** is the lighter, functional outer region. Rays (horizontal parenchyma strands) facilitate radial transport. Timber is classified as **diffuse-porous** (vessels uniform, e.g., maple) or **ring-porous** (large early vessels, e.g., oak).



Xylem: Water-Conducting Tissue

Xylem is a complex permanent tissue composed of four element types: **Tracheids** — elongated, tapering cells with lignified walls and pits; the sole conducting cells in gymnosperms and ferns. **Vessel elements** — shorter, wider cells joined end-to-end into continuous tubes (vessels); characteristic of angiosperms, allowing more efficient water transport. **Xylem fibers** — thick-walled cells providing mechanical support. **Xylem parenchyma** — living cells for storage and lateral transport. Protoxylem (first-formed, with annular/spiral thickenings) and metaxylem (later-formed, with reticulate/pitted thickenings) are distinguished by their maturation timing.



Phloem: Food-Conducting Tissue

Phloem transports organic nutrients (primarily sucrose) from sources to sinks. Its components include: **Sieve tube elements** — elongated, living cells lacking nuclei at maturity, connected by sieve plates; the main conducting cells in angiosperms. **Companion cells** — metabolically active cells closely associated with sieve tubes, controlling their function via plasmodesmata. **Phloem fibers** — sclerenchyma cells providing support (e.g., hemp, jute). **Phloem parenchyma** — for storage and lateral transport. Sieve cells (in gymnosperms) lack true sieve plates and companion cells, instead associating with albuminous cells.

Anomalous Secondary Growth

Anomalous secondary growth refers to deviations from the typical pattern of secondary growth produced by a single, continuous vascular cambium forming concentric rings of secondary xylem and phloem. These anomalies occur in many dicots and monocots and involve unusual cambial behavior, abnormal distribution of vascular bundles, or the formation of successive cambia. Such patterns are of particular interest because they represent evolutionary adaptations to specific ecological conditions, including water storage, climbing habits, or survival in arid environments.

Bignonia (Liana)

In *Bignonia*, the vascular cambium produces normal secondary tissues initially, but then develops **four wedge-shaped areas of phloem** embedded within the secondary xylem. These phloem wedges are formed because the cambium in certain regions produces phloem inward instead of xylem. This creates a characteristic cross-shaped pattern in transverse section, an adaptation thought to provide flexibility in the climbing stem while maintaining vascular continuity.

Boerhaavia (Nyctaginaceae)

Boerhaavia exhibits **successive cambia** — multiple concentric rings of cambium form sequentially. Each cambium produces a ring of vascular bundles embedded in conjunctive tissue. The outermost cambium is the most recently formed. This pattern results in multiple rings of vascular bundles rather than a single continuous cylinder of wood. This is an adaptation common in the Nyctaginaceae family and allows for continued radial expansion while maintaining functional vascular tissue.

Nyctanthes (Oleaceae)

In *Nyctanthes*, anomalous secondary growth involves the formation of **included phloem** — strands of phloem embedded within the secondary xylem. This occurs when portions of the cambium temporarily reverse their activity, producing phloem toward the center instead of xylem. The result is islands of phloem surrounded by xylem tissue, a pattern that may enhance the mechanical properties of the stem while maintaining transport capacity.

Key Types of Anomalies: (1) **Abnormal cambial behavior** — cambium produces unusual tissue patterns (e.g., phloem wedges in Bignonia); (2) **Successive cambia** — multiple cambial rings form sequentially (e.g., Boerhaavia, Beta vulgaris); (3) **Included phloem** — phloem strands embedded in xylem (e.g., Nyctanthes, Salvadora); (4) **Vascular bundles in ground tissue** — as in monocot stems like palms.

Microsporogenesis & Male Gametophyte

The male reproductive pathway in angiosperms begins within the anther, where specialized diploid cells undergo meiosis to produce haploid microspores, which then develop into the male gametophyte (pollen grain). This process is tightly regulated and involves precise cellular interactions between the developing pollen and the surrounding sporogenous and nutritive tissues of the anther.

Microsporogenesis

Microsporogenesis is the formation of haploid microspores from diploid microspore mother cells (pollen mother cells) through meiosis. The anther typically contains **four microsporangia** (pollen sacs), each lined by four wall layers: **epidermis** (protective), **endothecium** (aids in dehiscence via fibrous thickenings), **middle layers** (transient), and **tapetum** (innermost nutritive layer). The tapetum is critical for pollen development — it provides nutrients, enzymes, and materials for pollen wall formation (including sporopollenin for the exine).

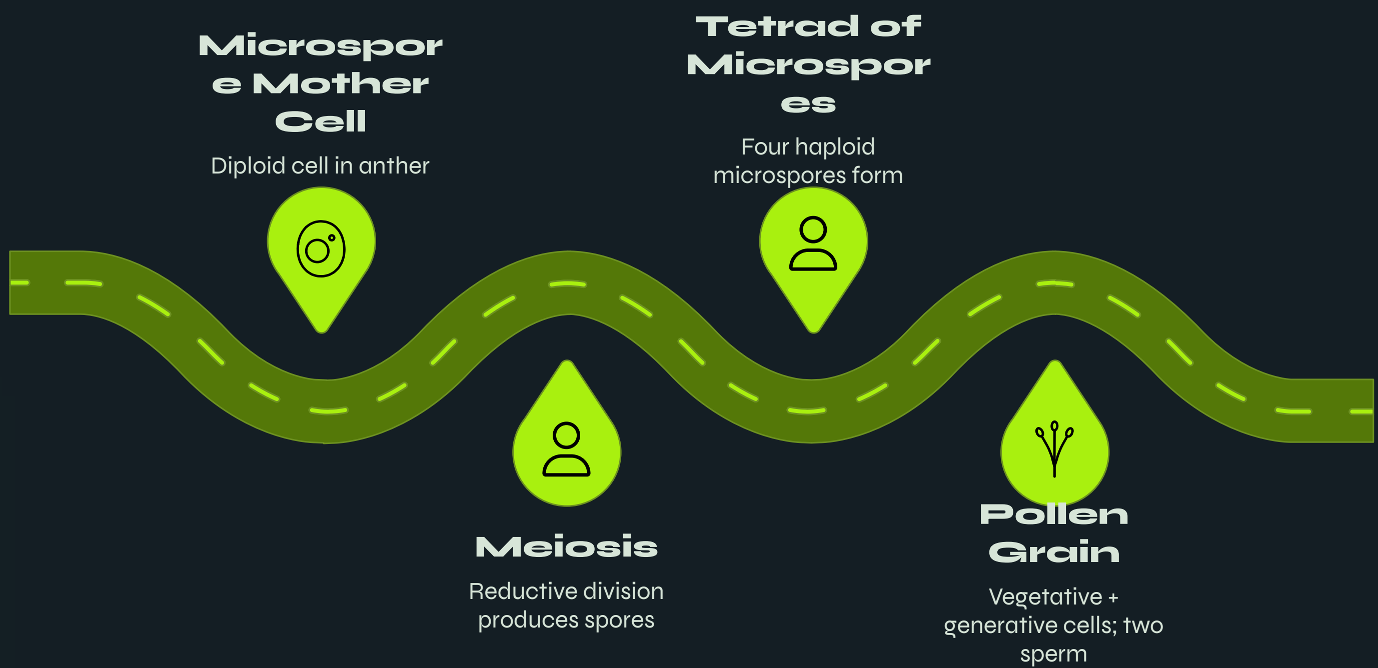
Each microspore mother cell undergoes meiosis to produce a **tetrad of four haploid microspores**. Tetrads may be tetrahedral, isobilateral, decussate, or T-shaped. Microspores are released as **monads** (most common), tetrads, or polyads depending on the species. The microspore then undergoes mitosis to form the immature male gametophyte.

Male Gametophyte Development

The microspore undergoes an unequal mitotic division to produce a large **vegetative cell** (tube cell) and a small **generative cell**. In many species, the generative cell divides again (either within the pollen grain or within the pollen tube) to form **two male gametes (sperm nuclei)**. The mature male gametophyte thus consists of three cells: one vegetative cell and two sperm cells.

The pollen grain wall has two layers: the outer **exine** (made of highly resistant sporopollenin, often sculptured with species-specific patterns) and the inner **intine** (cellulosic and pectic). Pollen grains may be shed at the 2-celled stage (vegetative + generative cell) or 3-celled stage (vegetative + two sperm cells), depending on the species. Upon landing on a compatible stigma, the pollen grain germinates and the **pollen tube** grows through the style, guided by chemical signals, to deliver the two sperm cells to the embryo sac.

Pollen grains are among the most resistant biological structures known — sporopollenin is chemically inert and preserves pollen in fossil records for millions of years, making palynology a powerful tool in paleobotany and archaeology.



The progression from diploid microspore mother cell to mature three-celled pollen grain involves one meiotic division followed by two mitotic divisions, reducing the chromosome number by half and producing the male gametes required for fertilization.

Megasporogenesis & Female Gametophyte

The female reproductive pathway in angiosperms occurs within the ovule, where a diploid megaspore mother cell undergoes meiosis to produce haploid megaspores, one of which develops into the female gametophyte (embryo sac). The embryo sac is the site of fertilization and the origin of both the embryo and endosperm. This process involves intricate cellular differentiation within the protected environment of the ovule.

Megasporogenesis

Megasporogenesis is the formation of haploid megaspores from a diploid megaspore mother cell (MMC) within the nucellus of the ovule. The MMC undergoes meiosis to produce a linear tetrad of four haploid megaspores. In the most common pattern (**Polygonum type**, monosporic development), only the **chalazal megaspore** remains functional while the other three degenerate. The functional megaspore then undergoes three successive mitotic divisions to form the eight-nucleate embryo sac.

Alternative patterns include **bisporic development** (two nuclei contribute, e.g., Allium type) and **tetrasporic development** (all four nuclei contribute, e.g., Fritillaria type), resulting in embryo sacs with different nuclear origins and arrangements.

The Female Gametophyte (Embryo Sac)

The typical angiosperm embryo sac (Polygonum type) is an **eight-nucleate, seven-celled structure** organized as follows:

Egg Apparatus (Micropylar End)

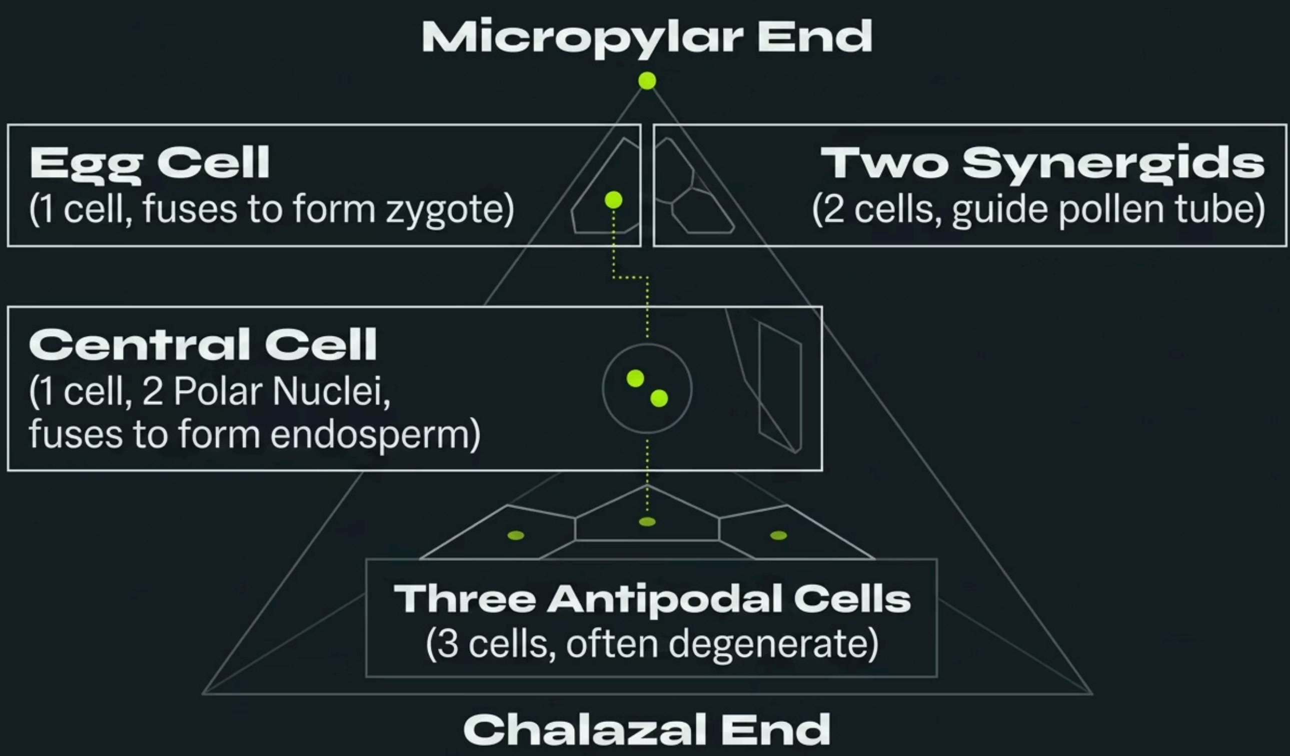
One **egg cell** (fertilized to form zygote) + two **synergids** (guide pollen tube via filiform apparatus; secrete attractants)

Central Cell

Contains two **polar nuclei** (fuse to form secondary nucleus; fertilized to form triploid endosperm)

Antipodal Cells (Chalazal End)

Three cells; often degenerate before or after fertilization; function varies by species



The embryo sac is highly polarized, with the micropylar end (egg apparatus) positioned to receive the pollen tube and the chalazal end (antipodals) oriented toward the base of the ovule. The central cell, occupying the largest volume, is the site of the second fertilization event that produces the nutritive endosperm.

Double Fertilization & Endosperm

Double fertilization is a unique and defining characteristic of angiosperms, discovered by Nawaschin in 1898. It involves two separate fertilization events occurring simultaneously within the embryo sac, ensuring that endosperm — the nutritive tissue — is only formed when fertilization is successful. This elegant mechanism prevents the wasteful production of endosperm in unfertilized ovules.

Double Fertilization

After the pollen tube enters the embryo sac (typically through a synergid, which degenerates in the process), it releases two sperm cells. The first sperm fuses with the **egg cell** to form a diploid (2n) **zygote** — this is **syngamy** or true fertilization. The second sperm fuses with the two **polar nuclei** (or the secondary nucleus) in the central cell to form a triploid (3n) **primary endosperm nucleus (PEN)** — this is **triple fusion**. Together, these two events constitute double fertilization.

The zygote undergoes a period of rest before dividing, while the primary endosperm nucleus typically divides immediately, ensuring that nutritive tissue is available for the developing embryo. This temporal coordination is critical for successful seed development.

Endosperm: Types & Development

The endosperm is the triploid nutritive tissue that nourishes the developing embryo. It shows three patterns of development:

Nuclear Type

Most common. Primary endosperm nucleus undergoes repeated free nuclear divisions without cell wall formation, creating a multinucleate coenocyte. Cell walls form later (centripetally). Examples: maize, coconut (liquid endosperm = free nuclei).

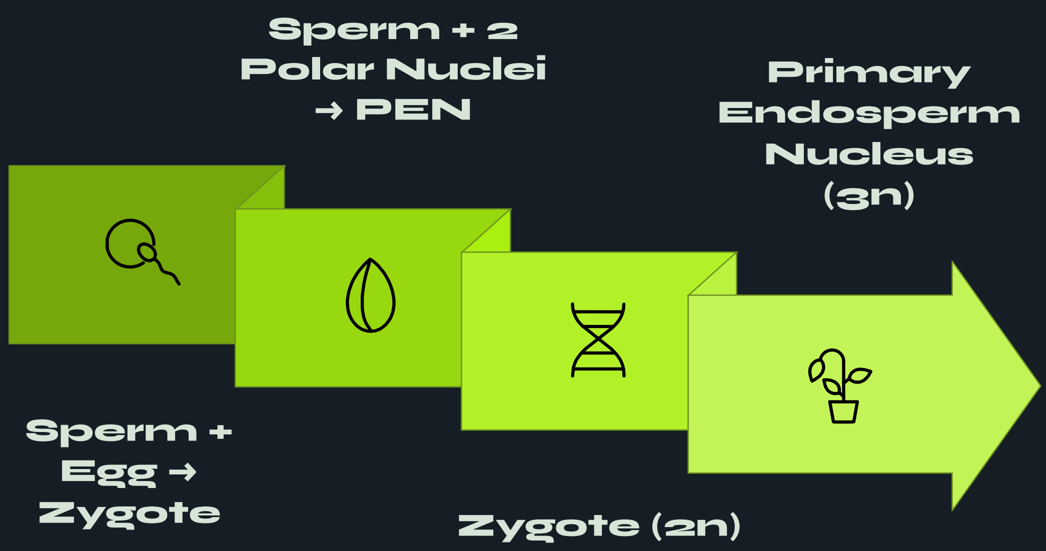
Cellular Type

Each nuclear division is immediately followed by cell wall formation. No free nuclear stage. Common in gamopetalous dicots. Examples: Datura, Petunia.

Helobial Type

Intermediate between nuclear and cellular. First division is followed by cell wall formation, creating two chambers; subsequent divisions are free nuclear. Common in monocots. Examples: Asphodelus.

Endosperm may be **consumed during embryo development** (non-endospermic/exalbuminous seeds, e.g., pea, bean) or **persist in the mature seed** (endospermic/albuminous seeds, e.g., maize, castor, coconut). In coconut, the liquid endosperm represents the free nuclear stage, while the solid kernel represents cellular endosperm.



The two simultaneous fertilization events ensure coordinated development of both the embryo and its nutritive tissue, a hallmark innovation of angiosperm reproduction that contributes to their extraordinary evolutionary success.

Embryogenesis

Embryogenesis is the process by which the zygote develops into a mature embryo within the seed. It involves a precisely orchestrated series of cell divisions and differentiation events that establish the basic body plan of the future plant, including the apical-basal axis (shoot-to-root polarity) and the radial axis (central vascular tissue surrounded by ground tissue and epidermis). Embryogenesis transforms a single cell into a multicellular organism with distinct organs and tissue systems.

1. Proembryo Stage

The zygote undergoes an asymmetric first division, producing a small **apical cell** (gives rise to most of the embryo) and a large **basal cell** (forms the suspensor). The suspensor anchors the embryo to the nucellus and facilitates nutrient transfer. The apical cell divides to form a globular mass of cells.

2. Globular Stage

The embryo is a spherical mass of cells. Cell divisions are relatively uniform. The **protoderm** (future epidermis) begins to differentiate at the surface. The embryo is radially symmetrical at this stage. In dicots, this stage establishes the foundation for subsequent organ differentiation.

3. Heart Stage

Cell divisions become localized, producing two lateral outgrowths — the **cotyledonary primordia** — giving the embryo a heart-shaped appearance in dicots. The **shoot apical meristem (SAM)** is established between the cotyledons. Vascular tissue differentiation begins. The apical-basal axis is clearly defined.

4. Torpedo Stage

The embryo elongates; cotyledons become prominent and elongated. The **root apical meristem (RAM)** differentiates at the basal end. The **hypocotyl** (region between cotyledons and radicle) is established. All major tissue systems are differentiated. The embryo is now structurally similar to a miniature plant.

5. Mature Embryo

The embryo reaches its final form: **plumule** (embryonic shoot with SAM), **two cotyledons** (dicots) or **one scutellum** (monocots), **hypocotyl**, and **radicle** (embryonic root with RAM). The embryo enters dormancy as the seed matures and desiccates, ready for germination under favorable conditions.

Monocot vs. Dicot Embryogenesis: In monocots (e.g., grasses), only one cotyledon (the **scutellum**) is formed. The embryonic shoot is enclosed in a **coleoptile** (protective sheath), and the radicle is covered by a **coleorhiza**. The scutellum functions in nutrient absorption from the endosperm during germination. In dicots, the two cotyledons may be photosynthetic (epigeal germination) or remain underground (hypogeal germination).

Seed Structure

A seed is a mature, fertilized ovule containing an embryo, stored food reserves, and a protective seed coat. It represents the culmination of the reproductive cycle and serves as the primary dispersal unit of seed plants. Seeds exhibit remarkable diversity in structure, size, and dormancy mechanisms, reflecting adaptations to different ecological niches and dispersal strategies.

Components of a Seed

Embryo

The living plant in miniature: **plumule** (embryonic shoot), **radicle** (embryonic root), **hypocotyl** (axis between), and **cotyledons** (seed leaves — two in dicots, one scutellum in monocots). Cotyledons may store food or absorb it from endosperm.

Seed Coat (Testa)

Derived from the ovule integuments. The outer integument forms the **testa**; the inner forms the **tegmen**. Provides mechanical protection, prevents desiccation, and may enforce dormancy. The **micropyle** (opening for water entry) and **hilum** (scar of attachment to funiculus) are visible features.

Stored Food

May be stored in **cotyledons** (exalbuminous seeds: bean, pea) or in **endosperm** (albuminous seeds: maize, castor, coconut). Reserves include starch, proteins (aleurone grains), and oils. In cereals, the **aleurone layer** secretes enzymes during germination to mobilize endosperm reserves.

Specialized Seed Features

Caruncle: A fleshy outgrowth near the micropyle, often oily and attractive to ants (myrmecochory). Found in *Ricinus* (castor) and *Euphorbia*. Aids in seed dispersal by ants.

Aril: A fleshy, often brightly colored covering that partially or wholly envelops the seed. Derived from the funiculus or ovule integuments. Attracts birds and mammals for dispersal. Examples: *Myristica* (nutmeg — mace is the aril), *Taxus*, *Acer*.

Seed Dormancy: A temporary suspension of germination despite favorable conditions. Types include: **physical dormancy** (hard seed coat prevents water uptake), **physiological dormancy** (embryo immaturity or inhibitors), **morphological dormancy** (undifferentiated embryo), and **combinational dormancy**. Dormancy ensures germination occurs at optimal times, preventing seedling mortality.

Monocot vs. Dicot Seeds: Dicot seeds (e.g., bean) have two large cotyledons that often fill the seed. Monocot seeds (e.g., maize grain) have a single scutellum, abundant endosperm, and protective coverings (coleoptile, coleorhiza). In maize, the fruit wall (pericarp) is fused with the seed coat, making it a **caryopsis**.

2n

Zygote Ploidy

Diploid embryo formed by fusion of haploid sperm and egg

3n

Endosperm Ploidy

Triploid nutritive tissue from double fertilization

7

Embryo Sac Cells

Seven cells, eight nuclei in the Polygonum-type embryo sac

4

Embryo Stages

Proembryo → Globular → Heart → Torpedo → Mature